

Nonparametric Statistical Analysis of Statewide Planning and Research Cooperative System 2022 Hospital Discharge Data: Kings and Queens Counties, New York

Vinay Siddi, Department of Data Science, University at Albany, SUNY

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1 Introduction

This project applies ten nonparametric statistical tests to the New York Statewide Planning and Research Cooperative System (SPARCS) 2022 hospital discharge dataset, restricted to Kings County (Brooklyn) and Queens County. The tests performed are the Runs Test, Sign Test, Wilcoxon Signed-Rank Test, Wilcoxon Rank Sum (Mann–Whitney U) Test, Kolmogorov–Smirnov Test, Kruskal–Wallis Test, Friedman Test, Kendall’s Tau, Spearman’s Rank Correlation, and the Permutation Test. The aim is to quantify, with distribution-free methods, how Length of Stay, Total Charges, Total Costs, and Birth Weight vary between the two counties and across demographic/clinical sub-groups.

2 Literature Review

A full literature review of prior works (1943–2024) motivating the use of distribution-free methods on hospital discharge data – presented in chronological order and including Swed and Eisenhart (1943); Efron and Tibshirani (1993); Ernst (2004); Fagerland and Sandvik (2009); Qualls, Pallin, and Schuur (2010); Croux and Dehon (2010); Anhoj and Olesen (2014); Moore et al. (2017); Chazard et al. (2017); Ning et al.

(2017); Bujang and Sapri (2018); Dieleman et al. (2020); Nguyen et al.

(2021); Stone et al. (2022); Li et al. (2022); and Heslin et al.

(2024) – is given in the accompanying LaTeX paper `Vinay_Final_Project.pdf`, Section 2. This R-Markdown file is the code companion and does not duplicate the full review.

3 Data Presentation and Exploratory Analysis

3.1 Load and Clean the Dataset

```
# Load the SPARCS 2022 Kings and Queens dataset
mydata <- read.csv("SPARCS_2022_Kings_Queens.csv")

# Derive four numeric columns from the raw (comma-formatted) SPARCS fields
```

```
mydata$Charges <- as.numeric(gsub(",", "", mydata$Total.Charges))
mydata$Costs <- as.numeric(gsub(",", "", mydata$Total.Costs))
mydata$LOS <- as.numeric(mydata$Length.of.Stay)
mydata$BirthWt <- as.numeric(mydata$Birth.Weight)
```

3.2 Dataset Dimensions and Structure

```
# Raw SPARCS CSV has 33 columns. We added 4 derived numeric columns above,
# so ncol(mydata) = 37 (33 original SPARCS + 4 derived).
cat("Number of Rows (Observations):", nrow(mydata), "\n")
```

```
## Number of Rows (Observations): 368495
```

```
cat("Number of Columns (Variables):", ncol(mydata),
    " (33 original SPARCS + 4 derived numeric)\n")
```

```
## Number of Columns (Variables): 37 (33 original SPARCS + 4 derived numeric)
```

```
cat("Size of Dataset (in-memory R object):",
    format(object.size(mydata), units = "MB"),
    "| CSV on disk: ~160 MB\n")
```

```
## Size of Dataset (in-memory R object): 136.9 Mb | CSV on disk: ~160 MB
```

```
cat("Time Period: Calendar Year 2022\n")
```

```
## Time Period: Calendar Year 2022
```

```
cat("Geographic Coverage: Kings County and Queens County, NYC\n")
```

```
## Geographic Coverage: Kings County and Queens County, NYC
```

```
cat("Counties:", paste(unique(mydata$Hospital.County), collapse = ", "), "\n")
```

```
## Counties: Queens, Kings
```

3.3 Variable Classification

```
# Note: this counts by R *storage type*, not by measurement scale.
# Several SPARCS fields (e.g., APR DRG Code, APR MDC Code, Operating
# Certificate Number, Permanent Facility ID) are stored as integers but
# are nominal classification labels - arithmetic on them is not
# meaningful. See the LaTeX paper Table 1 for the measurement-scale
# classification.
num_quant <- sum(sapply(mydata, is.numeric))
num_qual <- sum(sapply(mydata, function(x) is.character(x) || is.factor(x)))
cat("Numeric (R storage):", num_quant, "\n")
```

```
## Numeric (R storage): 10
```

```
cat("Character/factor (R storage):", num_qual, "\n")
```

```
## Character/factor (R storage): 27
```

By **measurement scale** (rather than R storage type), the 33 original SPARCS variables break down as: 4 truly quantitative (ratio-scale) variables — Length of Stay, Total Charges, Total Costs, Birth Weight; 1 ordinal variable — APR Severity of Illness Code (1–4); and 28 categorical (nominal) variables, including ID and classification codes (Operating Certificate Number, Permanent Facility ID, APR DRG Code, APR MDC Code) that are stored as integers but represent labels rather than measurements. Discharge Year is constant (2022) and contributes no variation.

3.4 Summary of Key Numeric Variables

```
summary(mydata[, c("LOS", "Charges", "Costs", "BirthWt")])
```

##	LOS	Charges	Costs	BirthWt
## Min. :	1.00	Min. : 7	Min. : 2	Min. : 400
## 1st Qu.:	2.00	1st Qu.: 21544	1st Qu.: 6746	1st Qu.:2900
## Median :	3.00	Median : 40497	Median : 12543	Median :3200
## Mean :	5.76	Mean : 71907	Mean : 23273	Mean :3159
## 3rd Qu.:	6.00	3rd Qu.: 77905	3rd Qu.: 24821	3rd Qu.:3500
## Max. :	119.00	Max. :12952195	Max. :5602076	Max. :6100
## NA's :	332			NA's :325132

3.5 Exploratory Graphs

The graphs below illustrate the distributional shape, skewness, and group differences that motivate the choice of nonparametric tests.

3.5.1 Boxplots of Length of Stay, Charges, and Costs by County

```
# Shared color palette for all county boxplots
county_cols <- c("#8FBFE0", "#F4A259")

# Length of Stay (in days) - log scale with plain day numbers
boxplot(LOS ~ Hospital.County,
        data = mydata[!is.na(mydata$LOS), ],
        log = "y", yaxt = "n",
        main = "Length of Stay by County (log scale)",
        ylab = "Length of Stay (days)",
        col = county_cols)
axis(2, at = c(1, 2, 5, 10, 20, 50, 100),
     labels = c("1", "2", "5", "10", "20", "50", "100"), las = 1)
```

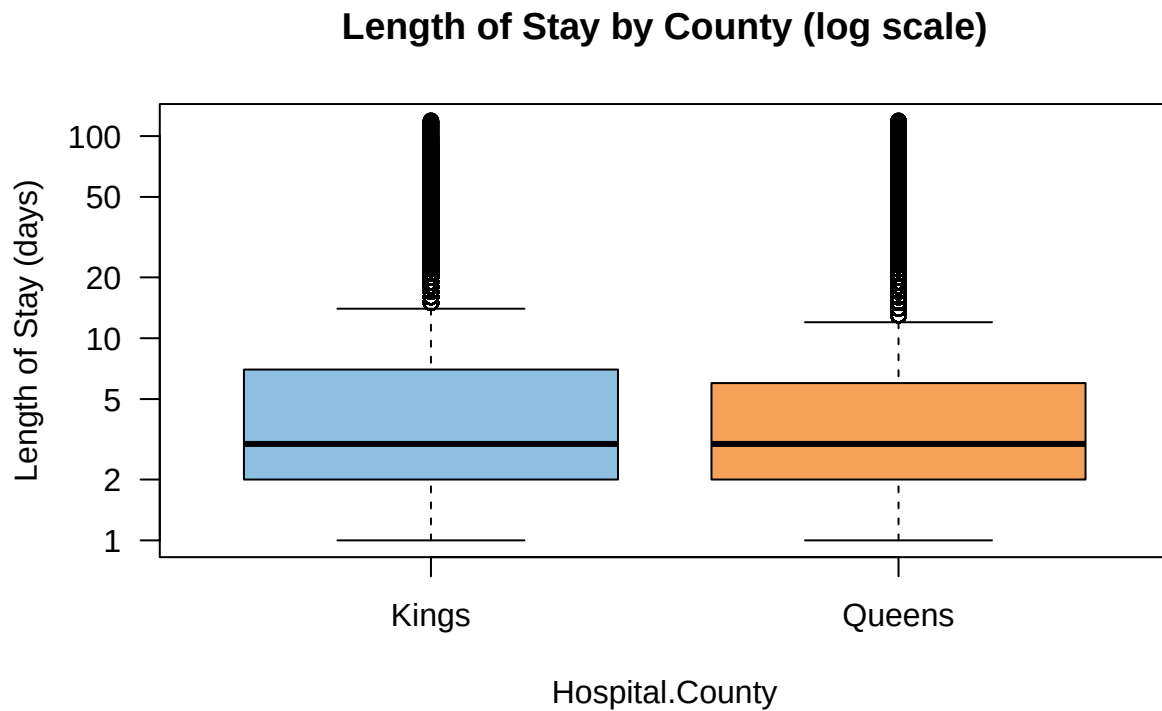


Figure 1: Log-scale boxplots by Hospital County.

```
# Total Charges - log scale with readable dollar labels
boxplot(Charges ~ Hospital.County,
  data = mydata[!is.na(mydata$Charges), ],
  log = "y", yaxt = "n",
  main = "Total Charges by County (log scale)",
  ylab = "Total Charges (USD)",
  col = county_cols)
axis(2, at = c(10, 100, 1000, 10000, 100000, 1000000, 10000000),
  labels = c("$10", "$100", "$1K", "$10K", "$100K", "$1M", "$10M"), las = 1)
```

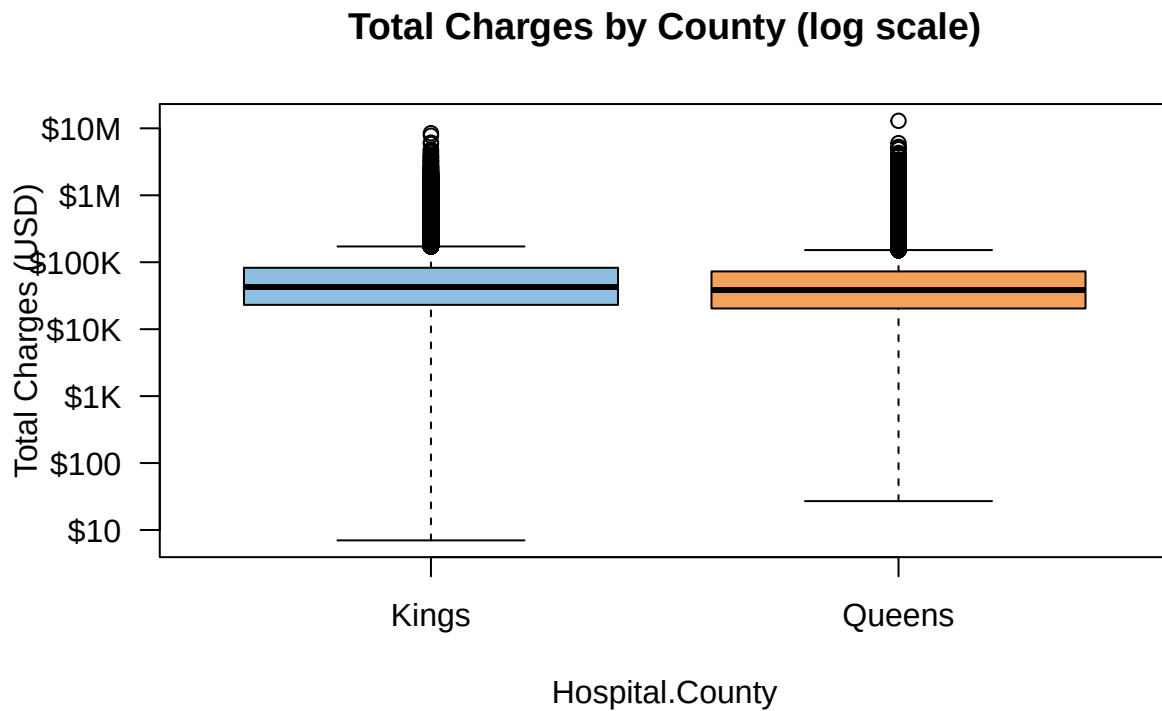


Figure 2: Log-scale boxplots by Hospital County.

```
# Total Costs - log scale with readable dollar labels
boxplot(Costs ~ Hospital.County,
        data = mydata[!is.na(mydata$Costs), ],
        log = "y", yaxt = "n",
        main = "Total Costs by County (log scale)",
        ylab = "Total Costs (USD)",
        col = county_cols)
axis(2, at = c(1, 10, 100, 1000, 10000, 100000, 1000000),
     labels = c("$1", "$10", "$100", "$1K", "$10K", "$100K", "$1M"), las = 1)
```

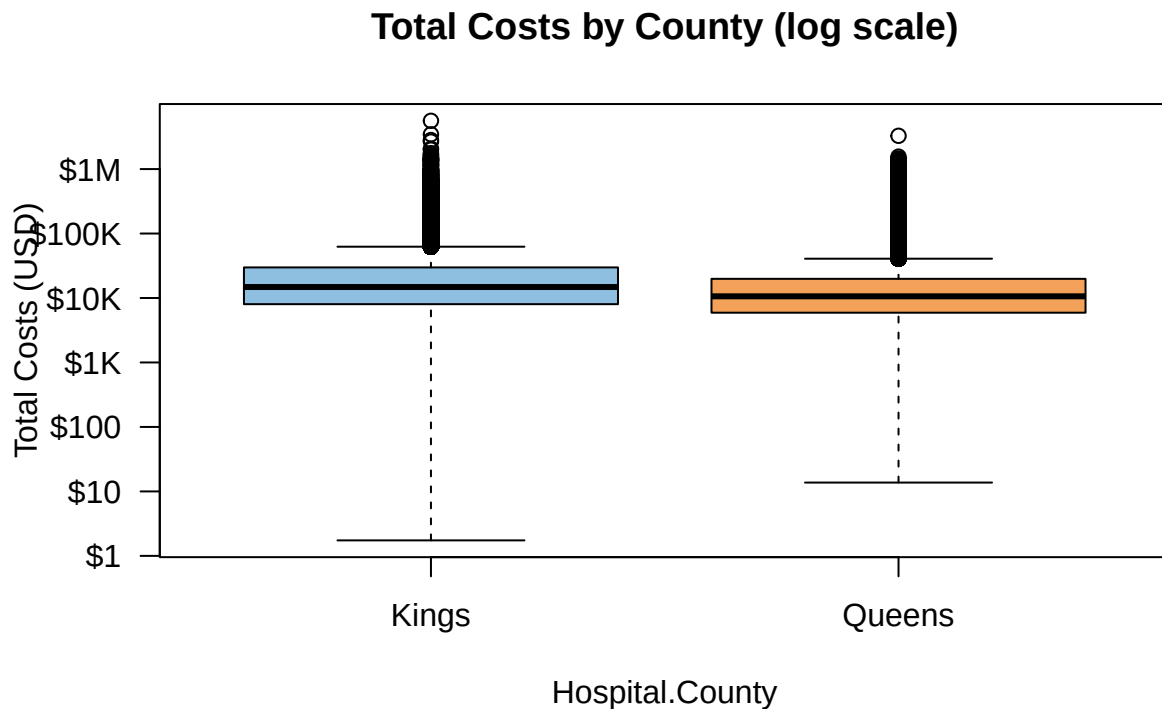
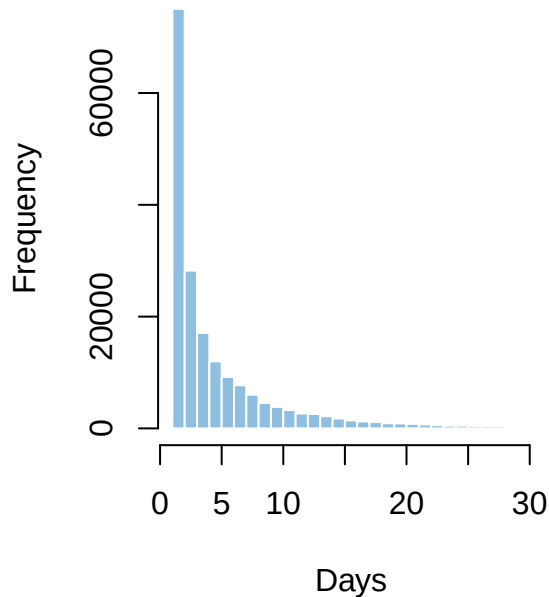
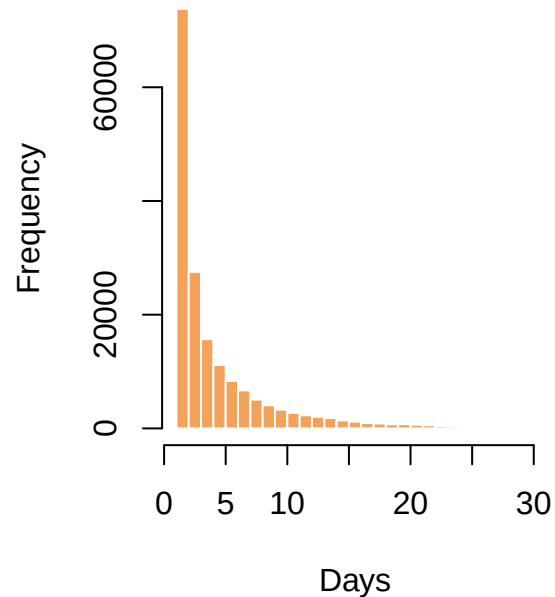


Figure 3: Log-scale boxplots by Hospital County.

What these plots show. The three boxplots above compare Length of Stay (in days), Total Charges (the dollar amount billed), and Total Costs (the actual cost to the hospital) between Kings and Queens counties. The thick black line in each box is the median (the typical patient), and the box itself covers the middle 50% of patients. The y-axis uses a log scale because values range from very small to very large; the small dots above each box are unusually long stays or expensive cases. In all three plots, Kings County sits slightly higher than Queens, meaning Kings patients tend to stay longer and incur higher charges and costs than Queens patients.

3.5.2 Histograms of Length of Stay (Capped at 30 Days)

```
par(mfrow = c(1, 2))
hist(mydata$LOS[mydata$Hospital.County == "Kings" & mydata$LOS <= 30],
     breaks = 30, col = county_cols[1], border = "white",
     main = "Kings: LOS (capped 30)", xlab = "Days")
hist(mydata$LOS[mydata$Hospital.County == "Queens" & mydata$LOS <= 30],
     breaks = 30, col = county_cols[2], border = "white",
     main = "Queens: LOS (capped 30)", xlab = "Days")
```

Kings: LOS (capped 30)**Queens: LOS (capped 30)**

```
par(mfrow = c(1, 1))
```

What these plots show. These two histograms show how often each length of stay occurs in each county, capped at 30 days so the shape stays readable. Both counties show the same pattern: a huge spike at 1 to 3 days (most patients go home quickly) and a long thin tail going out to 30 days (a small fraction stay much longer). This pattern is called “right-skewed” and is the reason this project uses nonparametric tests instead of standard tests that assume a normal bell curve. The two counties have nearly identical shapes, but Kings has slightly more discharges overall.

3.5.3 Bar Plot: Discharges by Age Group and County

```
barplot(t(table(mydata$Age.Group, mydata$Hospital.County)),
        beside = TRUE,
        col     = county_cols,
        legend.text = TRUE,
        args.legend = list(x = "topright", bty = "n"),
        main     = "Number of Discharges by Age Group and County",
        ylab     = "Number of Discharges",
        las     = 1)
```


Number of Discharges by Age Group and County

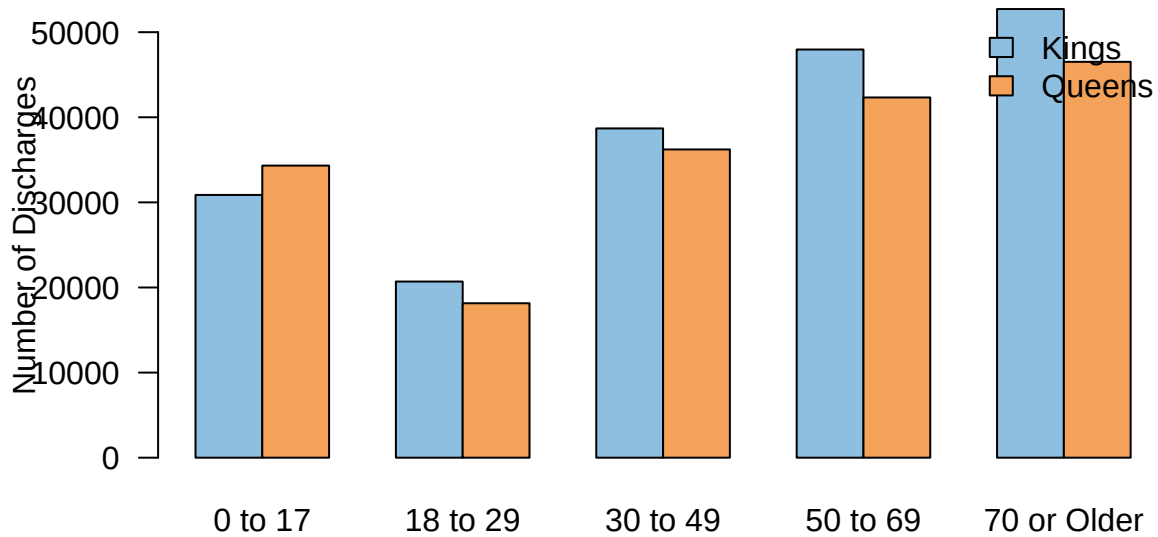


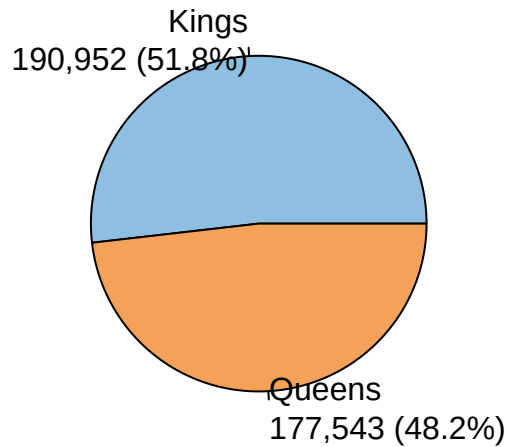
Figure 4: Discharges by Age Group and County.

What this plot shows. This bar chart counts how many hospital discharges fall into each age group, with Kings (blue) and Queens (orange) shown side by side. Both counties follow the same pattern: the elderly (70 or older) account for the most discharges, followed by adults aged 50 to 69. The fewest discharges are for young adults aged 18 to 29, who tend to be the healthiest age group. Kings County has more discharges than Queens in every adult age group, which matches its slightly larger total population.

3.5.4 Pie Chart: Share of Discharges by County

```
county_counts <- table(mydata$Hospital.County)
pie(county_counts,
  col = county_cols,
  main = "Share of Discharges: Kings vs Queens",
  labels = paste0(names(county_counts), "\n",
    format(county_counts, big.mark = ","),
    " (", round(100 * county_counts / sum(county_counts), 1), "%)")
```

Share of Discharges: Kings vs Queens



What this plot shows. This pie chart shows what share of the 368,495 hospital discharges in this dataset came from each county. Kings County (Brooklyn) accounts for slightly more than half (51.8%) and Queens accounts for the rest (48.2%). The split is fairly even, which means any group differences I find later in the paper come from real differences in patient populations and not just from one county having way more data than the other.

3.5.5 Empirical CDF of Total Charges

```
kings_ch <- mydata$Charges[mydata$Hospital.County == "Kings" & !is.na(mydata$Charges)]
queens_ch <- mydata$Charges[mydata$Hospital.County == "Queens" & !is.na(mydata$Charges)]

plot(ecdf(kings_ch), col = "#1F77B4", lwd = 2,
     main = "Empirical CDF of Total Charges",
     xlab = "Total Charges (USD)", xlim = c(0, 200000))
lines(ecdf(queens_ch), col = "#D62728", lwd = 2)
legend("bottomright", legend = c("Kings", "Queens"),
     col = c("#1F77B4", "#D62728"), lwd = 2, bty = "n")
```

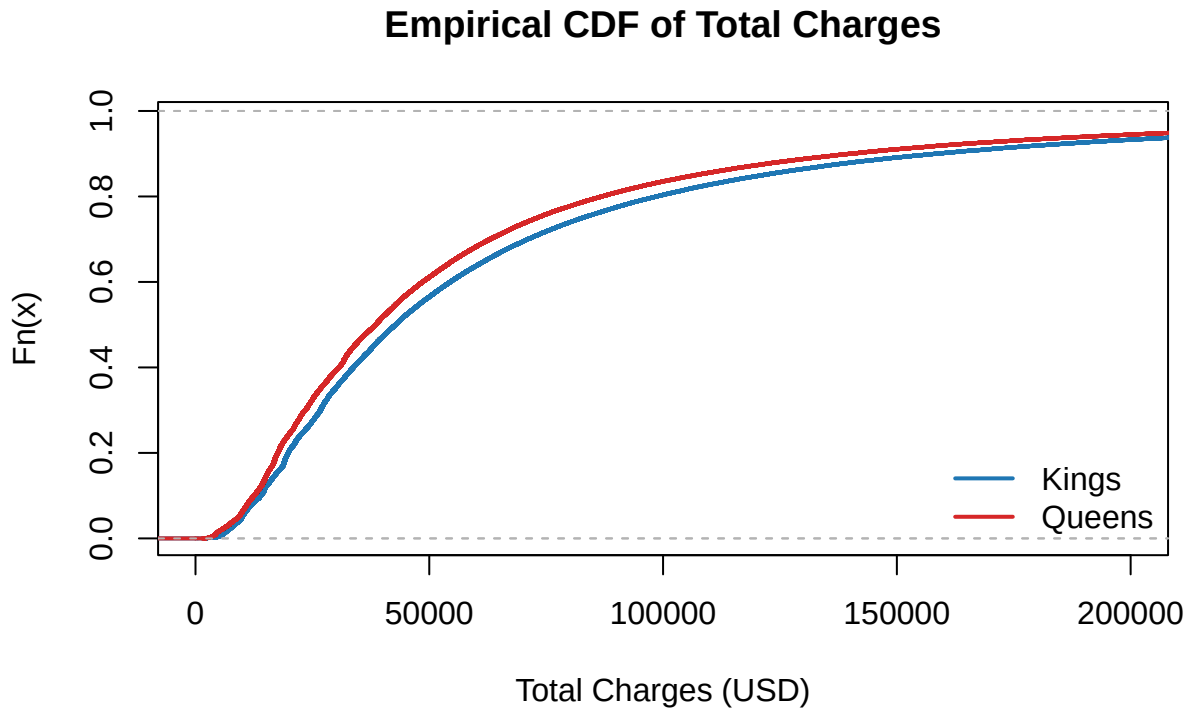


Figure 5: Empirical CDF of Total Charges by County.

What this plot shows. An empirical CDF (cumulative distribution function) shows what percentage of patients had charges at or below a given dollar amount. Reading the chart at \$50,000 on the x-axis, you can see that about 60% of Queens patients are charged at or below that amount, but only about 55% of Kings patients are. The Queens curve sits consistently above the Kings curve, meaning Queens patients tend to have lower charges across the board. The visible gap between the two curves is exactly what the Kolmogorov-Smirnov test detects to confirm the distributions are truly different.

3.5.6 Violin Plot of $\log_{10}(\text{Costs})$ by Severity

```
sev <- mydata[!is.na(mydata$Costs) &
              !is.na(mydata$APR.Severity.of.Illness.Code) &
              mydata$Costs > 0 & mydata$Costs < 200000 &
              mydata$APR.Severity.of.Illness.Code %in% 1:4, ]

sev_levels <- 1:4
sev_labels <- c("1-Minor", "2-Moderate", "3-Major", "4-Extreme")
pal        <- c("#BCE7FD", "#74C7EC", "#3282B8", "#0F4C81")
log_costs  <- log10(sev$Costs)

# Scaled kernel densities (one per severity level)
dens_list <- lapply(sev_levels, function(k) {
  d <- density(log_costs[sev$APR.Severity.of.Illness.Code == k], n = 512)
  d$y <- d$y / max(d$y) * 0.35
  d
})
```

```

})

plot(0, type = "n",
     xlim = c(0.5, 4.5), ylim = range(log_costs), xaxt = "n",
     main = "Violin Plot: log10(Total Costs) by APR Severity",
     xlab = "Severity Level", ylab = "log10(Total Costs)")
axis(1, at = 1:4, labels = sev_labels)

for (k in seq_along(sev_levels)) {
  d <- dens_list[[k]]
  vals <- log_costs[sev$APR.Severity.of.Illness.Code == sev_levels[k]]

  # Mirrored density polygon (violin)
  polygon(c(k + d$y, rev(k - d$y)), c(d$x, rev(d$x)),
         col = pal[k], border = "grey40")
  # Inner IQR box and median point
  rect(k - 0.06, quantile(vals, 0.25),
       k + 0.06, quantile(vals, 0.75),
       col = "grey30", border = NA)
  points(k, median(vals), pch = 21, bg = "white", cex = 1.2)
}

```

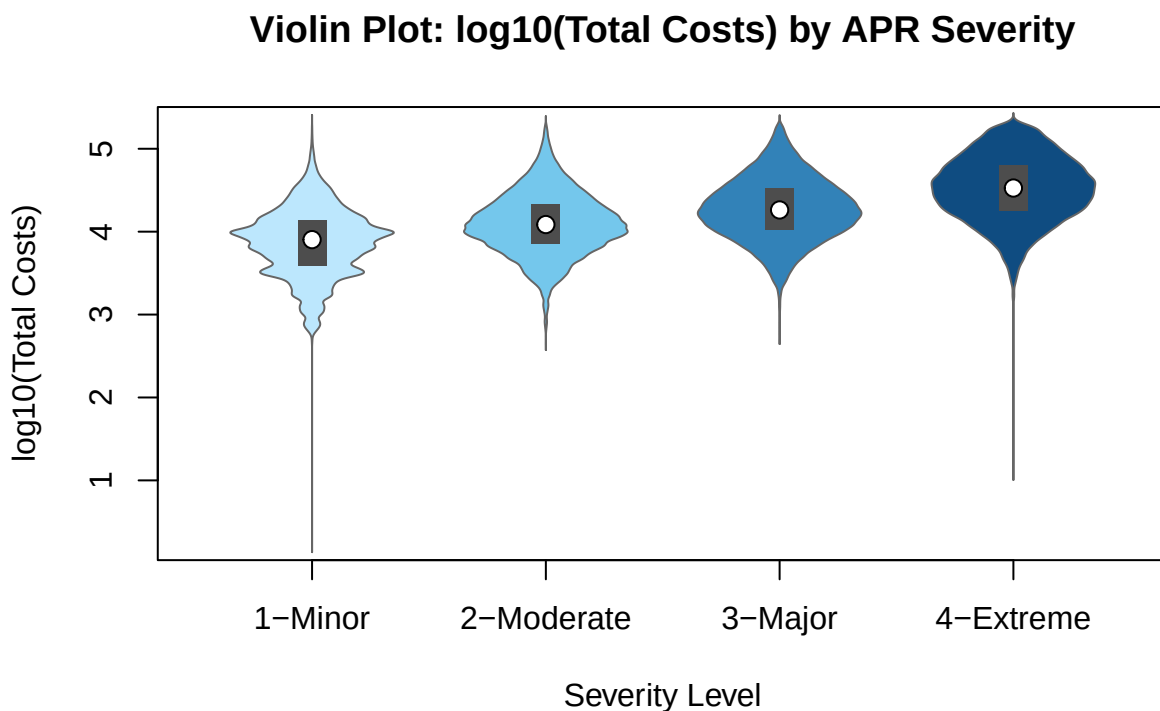


Figure 6: Violin plot of log10(Total Costs) by APR Severity of Illness.

What this plot shows. A violin plot is like a boxplot but with the box “fattened” to show how common each cost level is. Each violin represents one severity level, from Minor (light blue)

on the left to Extreme (dark blue) on the right. The wider sections show where most patients in that severity group have their costs. As severity worsens from Minor to Extreme, the violins shift upward (so costs go up) and get wider at the top (so the most expensive cases get even more expensive). This confirms that sicker patients cost the hospital more, in a smooth and predictable way.

3.5.7 Dot Plot: Median Total Charges by Age Group

```
age_medians <- sort(tapply(mydata$Charges, mydata$Age.Group, median, na.rm = TRUE))

dotchart(age_medians / 1000,
  labels = names(age_medians),
  main = "Median Total Charges by Age Group",
  xlab = "Median Total Charges (USD thousands)",
  pch = 19, color = "#1F77B4", cex = 1.1)
abline(v = seq(0, 150, by = 10), col = "grey85", lty = 2)
```

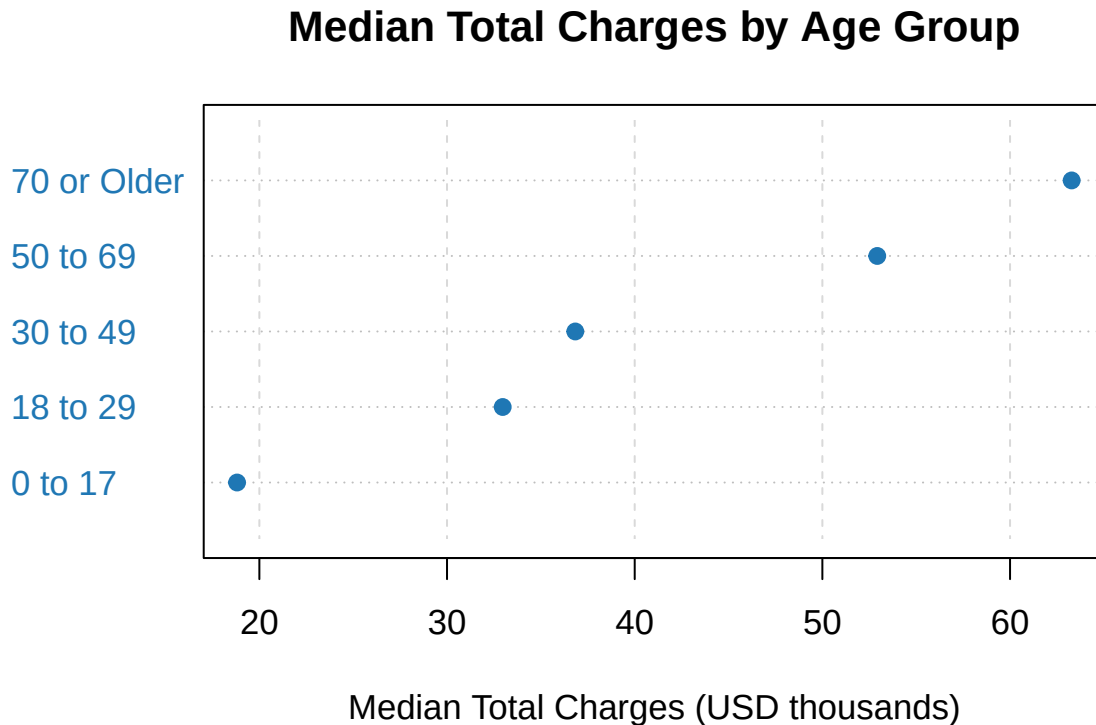


Figure 7: Dot plot of median Total Charges by Age Group.

What this plot shows. This dot plot shows the median (typical) hospital charge for each age group, in thousands of dollars. The pattern is very clear: the older the patient, the higher the typical charge. A 70-year-old patient's typical hospital bill (about \$63,000) is more than three times higher than a child's (about \$20,000). This makes intuitive sense because older patients tend to have more complex health conditions that need longer stays and more procedures. The Kruskal-Wallis test confirms this trend is statistically significant.

4 Objective Questions and Research Questions

Each row below gives one objective question (verbal form) and the corresponding null hypothesis at the test level. The full per-variable table – with separate objective question, research question, null hypothesis, test statistic, p-value, decision, and interpretation for each individual variable tested under each procedure – is provided in the accompanying LaTeX paper `Vinay_Final_Project.pdf`, Table~4. Significance level $\alpha = 0.05$ is used throughout.

Table 1: Research questions and null hypotheses

Test	Objective Question	Null Hypothesis
Runs Test	Are quantitative sequences random?	H_0 : sequence is random
Sign Test	Do medians differ between Kings and Queens (paired)?	$H_0: \tilde{\mu}_K = \tilde{\mu}_Q$
Wilcoxon Signed-Rank	Do paired distributions differ, accounting for magnitude?	H_0 : median of paired differences = 0
Mann–Whitney U	Do independent distributions differ between Kings and Queens / ED vs Non-ED?	H_0 : distributions equal between groups
Kolmogorov–Smirnov	Do full CDFs of LOS/Charges/Costs/BirthWt differ between Kings and Queens?	$H_0: F_K(x) = F_Q(x);$ $F_{ED}(x) = F_{Non-ED}(x)$
Kruskal–Wallis	Do distributions differ across Age Group, Severity, Admission, Payer?	$H_0: F_1 = F_2 = \dots = F_k$
Friedman	Do distributions differ across Severity levels, blocking by facility?	H_0 : no treatment effect within blocks
Kendall’s Tau	Is there ordinal association between pairs of numeric variables?	$H_0: \tau = 0$
Spearman’s Rho	Is there monotone rank association between pairs of numeric variables?	$H_0: \rho_s = 0$
Permutation Test	Is the median difference of Charges between counties attributable to chance?	H_0 : label is exchangeable

5 Nonparametric Analyses

5.1 Helper: Consistent Decision Label

```
# One-line helper used throughout the test sections so every table reports
# "Reject H0" / "Fail to Reject H0" the same way.
decide <- function(p, alpha = 0.05) {
  ifelse(is.na(p) | p >= alpha, "Fail to Reject H0", "Reject H0")
}
```

5.2 Test 1: Runs Test for Randomness

$$Z = \frac{R - E(R)}{\sqrt{\text{Var}(R)}}, \quad E(R) = \frac{2n_1n_2}{n} + 1$$

```
# Custom implementation (base R only): dichotomize around the median,
# count runs, and use the normal approximation for large samples.
runs_test <- function(x) {
  x <- na.omit(x)
  s <- ifelse(x > median(x), 1, 0)
  R <- 1 + sum(diff(s) != 0)
  n1 <- sum(s == 1); n2 <- sum(s == 0); n <- n1 + n2
  ER <- 2 * n1 * n2 / n + 1
  VarR <- 2 * n1 * n2 * (2 * n1 * n2 - n) / (n^2 * (n - 1))
  Z <- (R - ER) / sqrt(VarR)
  list(runs = R, expected = ER, z = Z,
       p = 2 * (1 - pnorm(abs(Z))))
}

runs_vars <- c("LOS", "Charges", "Costs", "BirthWt",
               "APR.DRG.Code", "APR.MDC.Code",
               "APR.Severity.of.Illness.Code")

runs_df <- do.call(rbind, lapply(runs_vars, function(v) {
  r <- runs_test(mydata[[v]])
  data.frame(Variable = v,
             Observed_Runs = r$runs,
             Expected_Runs = round(r$expected, 2),
             Z = round(r$z, 4),
             P_value = signif(r$p, 4),
             Decision = decide(r$p),
             stringsAsFactors = FALSE)
}))

knitr::kable(runs_df, caption = "Runs Test Results", row.names = FALSE)
```

Table 2: Runs Test Results

Variable	Observed_Runs	Expected_Runs	Z	P_value	Decision
LOS	164962	181790.51	-56.1691	0.00000	Reject H0
Charges	156725	184248.50	-90.6814	0.00000	Reject H0
Costs	161381	184248.50	-75.3414	0.00000	Reject H0
BirthWt	21248	21452.01	-1.9804	0.04766	Reject H0
APR.DRG.Code	168061	184232.92	-53.2860	0.00000	Reject H0
APR.MDC.Code	167813	184205.96	-54.0222	0.00000	Reject H0
APR.Severity.of.Illness.Code	152218	161003.40	-33.1242	0.00000	Reject H0

Interpretation. At $\alpha = 0.05$ all testable variables exhibit a non-random ordering. This is expected: SPARCS records are grouped by facility and admission date, so the sequence of values carries facility-level structure.

Limitation note on nominal codes. The Runs Test technically requires an ordered numeric sequence so that “above the median” and “below the median” carry meaning. Two of the seven variables tested above (`APR.DRG.Code` and `APR.MDC.Code`) are nominal classification labels stored as integers, so dichotomising them around their numeric median is mathematically valid but not clinically interpretable – the rejection of H_0 for those two rows reflects only that the file is sorted by facility, not a clinical pattern. The four genuinely quantitative variables (Length of Stay, Total Charges, Total Costs, Birth Weight) plus the ordinal APR Severity of Illness Code give the substantively meaningful Runs Test results.

5.3 Paired-Data Setup (Tests 2 and 3)

Tests 2 and 3 both compare Kings vs Queens on the same five quantitative variables. The block below builds the paired samples **once** (same seed throughout) and re-uses them.

```
paired_vars <- c("LOS", "Charges", "Costs", "BirthWt",
  "APR.Severity.of.Illness.Code")
paired_lbl <- c("Length of Stay", "Total Charges", "Total Costs",
  "Birth Weight", "APR Severity of Illness")

# Build one paired (Kings, Queens) sample per variable. We use the same
# seed (42) every time so the Sign Test and the Wilcoxon Signed-Rank Test
# operate on exactly the same pairs.
make_pair <- function(v) {
  k <- mydata[mydata$Hospital.County == "Kings" & !is.na(mydata[[v]]), v]
  q <- mydata[mydata$Hospital.County == "Queens" & !is.na(mydata[[v]]), v]
  n <- min(length(k), length(q))
  set.seed(42)
  list(kings = sample(k, n), queens = sample(q, n))
}

paired_samples <- setNames(lapply(paired_vars, make_pair), paired_vars)
```

5.4 Test 2: Sign Test

$$H_0 : \tilde{\mu}_K = \tilde{\mu}_Q, \quad n^+ \sim \text{Binomial}(n, 0.5) \text{ under } H_0$$

```
# Sign test on a single pair of aligned vectors
run_sign_test <- function(x, y) {
  d <- (x - y)[!is.na(x - y)]
  d <- d[d != 0] # drop ties
  n <- length(d); np <- sum(d > 0)
  if (n == 0) return(list(S = NA, p = NA))
  list(S = np, p = binom.test(np, n, 0.5)$p.value)
}
```



```

sign_results <- do.call(rbind, lapply(seq_along(paired_vars), function(i) {
  ps <- paired_samples[[i]]
  r <- run_sign_test(ps$Kings, ps$Queens)
  data.frame(Variable = paired_lbl[i],
             S       = r$S,
             P_value = signif(r$p, 4),
             Decision = decide(r$p),
             stringsAsFactors = FALSE)
}))

knitr::kable(sign_results,
  caption = "Sign Test Results: Kings vs Queens (all 5 paired variables)",
  row.names = FALSE)

```

Table 3: Sign Test Results: Kings vs Queens (all 5 paired variables)

Variable	S	P_value	Decision
Length of Stay	81301	0	Reject H0
Total Charges	94864	0	Reject H0
Total Costs	105522	0	Reject H0
Birth Weight	10374	0	Reject H0
APR Severity of Illness	65885	0	Reject H0

5.5 Test 3: Wilcoxon Signed-Rank Test

$$V = \min(T^+, T^-)$$

```

wsr_results <- do.call(rbind, lapply(seq_along(paired_vars), function(i) {
  ps <- paired_samples[[i]]
  wt <- suppressWarnings(wilcox.test(ps$Kings, ps$Queens, paired = TRUE))
  data.frame(Variable = paired_lbl[i],
             V       = as.numeric(wt$statistic),
             P_value = signif(wt$p.value, 4),
             Decision = decide(wt$p.value),
             stringsAsFactors = FALSE)
}))

knitr::kable(wsr_results,
  caption = "Wilcoxon Signed-Rank Test Results: Kings vs Queens (all 5 paired variables)",
  row.names = FALSE)

```

Table 4: Wilcoxon Signed-Rank Test Results: Kings vs Queens (all 5 paired variables)

Variable	V	P_value	Decision
Length of Stay	6399613688	0	Reject H0
Total Charges	8591237270	0	Reject H0
Total Costs	9924918650	0	Reject H0
Birth Weight	103797764	0	Reject H0
APR Severity of Illness	4172375010	0	Reject H0

5.5.1 Combined Paired Tests Summary

```
paired_combined <- merge(
  sign_results[, c("Variable", "P_value", "Decision")],
  wsr_results[, c("Variable", "P_value", "Decision")],
  by = "Variable", suffixes = c("_Sign", "_WSR"))

knitr::kable(paired_combined,
  caption = "Combined Paired Tests Summary (Sign Test vs Wilcoxon Signed-Rank)",
  col.names = c("Variable", "P-value (Sign)", "Decision (Sign)",
    "P-value (WSR)", "Decision (WSR)"),
  row.names = FALSE)
```

Table 5: Combined Paired Tests Summary (Sign Test vs Wilcoxon Signed-Rank)

Variable	P-value (Sign)	Decision (Sign)	P-value (WSR)	Decision (WSR)
APR Severity of Illness	0	Reject H0	0	Reject H0
Birth Weight	0	Reject H0	0	Reject H0
Length of Stay	0	Reject H0	0	Reject H0
Total Charges	0	Reject H0	0	Reject H0
Total Costs	0	Reject H0	0	Reject H0

5.6 Test 4: Wilcoxon Rank Sum Test (Mann–Whitney U Test)

```
# Each row: one two-sample comparison, variable ~ grouping.
mw_specs <- list(
  list(lbl = "Total Charges: Kings vs Queens",
    f = Charges ~ Hospital.County, na = "Charges"),
  list(lbl = "Total Costs: Kings vs Queens",
```

```

      f = Costs ~ Hospital.County,          na = "Costs"),
list(lbl = "Length of Stay: Kings vs Queens",
      f = LOS ~ Hospital.County,          na = "LOS"),
list(lbl = "Total Charges: ED vs Non-ED",
      f = Charges ~ Emergency.Department.Indicator,
      na = "Charges"),
list(lbl = "Birth Weight: Kings vs Queens",
      f = BirthWt ~ Hospital.County,      na = "BirthWt"),
list(lbl = "APR Severity: Kings vs Queens",
      f = APR.Severity.of.Illness.Code ~ Hospital.County,
      na = "APR.Severity.of.Illness.Code"))

mw_results <- do.call(rbind, lapply(mw_specs, function(s) {
  d <- mydata[!is.na(mydata[[s$na]]), ]
  mw <- wilcox.test(s$f, data = d)
  data.frame(Comparison = s$lbl,
             W           = as.numeric(mw$statistic),
             P_value     = signif(mw$p.value, 4),
             Decision    = decide(mw$p.value),
             stringsAsFactors = FALSE)
}))

knitr::kable(mw_results,
  caption = "Mann-Whitney U Test Results (Independent Samples)",
  row.names = FALSE)

```

Table 6: Mann-Whitney U Test Results (Independent Samples)

Comparison	W	P_value	Decision
Total Charges: Kings vs Queens	18127110461	0	Reject H0
Total Costs: Kings vs Queens	20174267488	0	Reject H0
Length of Stay: Kings vs Queens	17646194676	0	Reject H0
Total Charges: ED vs Non-ED	10784366186	0	Reject H0
Birth Weight: Kings vs Queens	248664478	0	Reject H0
APR Severity: Kings vs Queens	17498146870	0	Reject H0

5.7 Test 5: Kolmogorov–Smirnov Two-Sample Test

$$D = \sup_x |F_1(x) - F_2(x)|, \quad Z = D \sqrt{\frac{n_1 n_2}{n_1 + n_2}}$$

*# Reusable helper: run two-sample KS on a list of variables between two
subsets of mydata (defined by a grouping variable and its two levels).*

```

ks_compare <- function(group_var, lvl1, lvl2, vars, labels) {
  d1 <- mydata[mydata[[group_var]] == lvl1, ]
  d2 <- mydata[mydata[[group_var]] == lvl2, ]
  do.call(rbind, lapply(seq_along(vars), function(i) {
    x <- na.omit(d1[[vars[i]]]); y <- na.omit(d2[[vars[i]]])
    if (length(x) <= 5 || length(y) <= 5) return(NULL)
    kr <- ks.test(x, y)
    data.frame(Variable = labels[i],
               D       = round(as.numeric(kr$statistic), 5),
               P_value = signif(kr$p.value, 4),
               Decision = decide(kr$p.value),
               stringsAsFactors = FALSE)
  )))
}

ks_vars <- c("LOS", "Charges", "Costs", "BirthWt")
ks_labels <- c("Length of Stay", "Total Charges",
               "Total Costs", "Birth Weight")

```

5.7.1 Kings vs Queens

```

ks_kq <- ks_compare("Hospital.County", "Kings", "Queens",
                    ks_vars, ks_labels)
knitr::kable(ks_kq,
  caption = "KS Two-Sample Test: Kings vs Queens",
  row.names = FALSE)

```

Table 7: KS Two-Sample Test: Kings vs Queens

Variable	D	P_value	Decision
Length of Stay	0.02984	0	Reject H0
Total Charges	0.05510	0	Reject H0
Total Costs	0.14394	0	Reject H0
Birth Weight	0.04642	0	Reject H0

5.7.2 Emergency vs Non-Emergency Admissions

```

ks_ed <- ks_compare("Emergency.Department.Indicator", "Y", "N",
                    ks_vars, ks_labels)
knitr::kable(ks_ed,
  caption = "KS Two-Sample Test: Emergency vs Non-Emergency",
  row.names = FALSE)

```

Table 8: KS Two-Sample Test: Emergency vs Non-Emergency

Variable	D	P_value	Decision
Length of Stay	0.29586	0.000000	Reject H0
Total Charges	0.19504	0.000000	Reject H0
Total Costs	0.16739	0.000000	Reject H0
Birth Weight	0.05297	0.001553	Reject H0

5.8 Test 6: Kruskal–Wallis Test

$$H = \frac{12}{N(N+1)} \sum_{i=1}^k \frac{R_i^2}{n_i} - 3(N+1)$$

```

kw_groups <- c("Age.Group", "APR.Severity.of.Illness.Code",
              "Type.of.Admission", "Payment.Typology.1")
kw_labels <- c("Age Group", "APR Severity",
              "Type of Admission", "Primary Payer")

# Every outcome x every grouping (skip combinations with < 2 levels)
kw_df <- do.call(rbind, lapply(seq_along(ks_vars), function(i) {
  v <- ks_vars[i]
  do.call(rbind, lapply(seq_along(kw_groups), function(j) {
    g <- kw_groups[j]
    d <- mydata[!is.na(mydata[[v]]) & mydata[[g]] != "" & !is.na(mydata[[g]]), ]
    if (length(unique(d[[g]])) < 2) return(NULL)
    kw <- kruskal.test(as.formula(paste(v, "~", g)), data = d)
    data.frame(Variable = ks_labels[i],
               Grouping = kw_labels[j],
               Chi_Sq   = round(as.numeric(kw$statistic), 3),
               DF       = as.numeric(kw$parameter),
               P_value  = signif(kw$p.value, 4),
               Decision = decide(kw$p.value),
               stringsAsFactors = FALSE)
  })
}))

knitr::kable(kw_df, caption = "Kruskal--Wallis Test Results",
             row.names = FALSE)

```

Table 9: Kruskal–Wallis Test Results

Variable	Grouping	Chi_Sq	DF	P_value	Decision
Length of Stay	Age Group	39479.202	4	0	Reject H0
Length of Stay	APR Severity	82402.339	4	0	Reject H0
Length of Stay	Type of Admission	25197.769	5	0	Reject H0
Length of Stay	Primary Payer	28581.292	8	0	Reject H0
Total Charges	Age Group	65223.280	4	0	Reject H0
Total Charges	APR Severity	78528.195	4	0	Reject H0
Total Charges	Type of Admission	45557.536	5	0	Reject H0
Total Charges	Primary Payer	35086.832	8	0	Reject H0
Total Costs	Age Group	61784.571	4	0	Reject H0
Total Costs	APR Severity	76946.788	4	0	Reject H0
Total Costs	Type of Admission	50004.526	5	0	Reject H0
Total Costs	Primary Payer	25577.709	8	0	Reject H0
Birth Weight	APR Severity	1561.709	4	0	Reject H0
Birth Weight	Type of Admission	40.473	4	0	Reject H0
Birth Weight	Primary Payer	52.596	7	0	Reject H0

Note. Birth Weight $\times$ Age Group is degenerate: Birth Weight is non-missing only for newborn discharges, which all fall in the “0 to 17” age group, so only one level of Age Group has data and Kruskal–Wallis cannot be computed. All 15 remaining combinations reject H_0 at $\alpha = 0.05$.

5.9 Test 7: Friedman Test (Facility as Block, Severity as Treatment)

$$Q = \frac{12}{nk(k+1)} \sum_{j=1}^k R_j^2 - 3n(k+1)$$

```
# Severity (k=4 levels) is the treatment; facility is the block.
# Only facilities that have observations at all 4 severity levels are kept.
fried_df <- do.call(rbind, lapply(seq_len(3), function(i) {
  v <- ks_vars[i]
  sub <- mydata[!is.na(mydata[[v]]) &
    !is.na(mydata$APR.Severity.of.Illness.Code), ]

  ag <- aggregate(as.formula(paste(v,
    "~ Facility.Name + APR.Severity.of.Illness.Code")),
    data = sub, FUN = median)

  # Keep only facilities observed at all 4 severity levels
  complete <- names(table(ag$Facility.Name)[table(ag$Facility.Name) == 4])
  ag2 <- ag[ag$Facility.Name %in% complete, ]
```

```

if (length(complete) < 2) return(NULL)

# Wide format: rows = facilities (blocks), cols = severity levels (treatments)
w <- reshape(ag2, idvar = "Facility.Name",
             timevar = "APR.Severity.of.Illness.Code",
             direction = "wide")
m <- as.matrix(w[, -1])
fr <- friedman.test(m)

data.frame(Variable = ks_labels[i],
           N_Blocks = nrow(m),
           K         = ncol(m),
           Chi_Sq    = round(as.numeric(fr$statistic), 3),
           DF        = as.numeric(fr$parameter),
           P_value   = signif(fr$p.value, 4),
           Decision  = decide(fr$p.value),
           stringsAsFactors = FALSE)
)))

knitr::kable(fried_df,
             caption = "Friedman Test: Severity as Treatment, Facility as Block",
             row.names = FALSE)

```

Table 10: Friedman Test: Severity as Treatment, Facility as Block

Variable	N_Blocks	K	Chi_Sq	DF	P_value	Decision
Length of Stay	5	4	13.286	3	0.004058	Reject H0
Total Charges	5	4	15.000	3	0.001817	Reject H0
Total Costs	5	4	15.000	3	0.001817	Reject H0

5.10 Rank-Correlation Setup (Tests 8 and 9)

The Kendall and Spearman tests both loop over the same 6 variable pairs. Kendall's Tau is computed on a random 5,000-row subsample because exact computation on the full $N \approx 370,000$ records is infeasible.

```

num_vars <- c("LOS", "Charges", "Costs", "BirthWt")

# All unique (i,j) variable pairs, i < j
pair_idx <- t(combn(length(num_vars), 2))

# Stratified subsample for Kendall's Tau
set.seed(1)
sub <- mydata[sample(nrow(mydata), 5000), ]

```

5.11 Test 8: Kendall's Tau

$$\tau = \frac{C - D}{\binom{n}{2}}$$

```
kendall_df <- do.call(rbind, apply(pair_idx, 1, function(p) {
  v1 <- num_vars[p[1]]; v2 <- num_vars[p[2]]
  x <- sub[[v1]]; y <- sub[[v2]]
  ok <- complete.cases(x, y)
  if (sum(ok) <= 20) return(NULL)
  kt <- cor.test(x[ok], y[ok], method = "kendall", exact = FALSE)
  data.frame(Pair      = paste(v1, "vs", v2),
             Tau       = round(as.numeric(kt$estimate), 4),
             P_value   = signif(kt$p.value, 4),
             Decision  = decide(kt$p.value),
             stringsAsFactors = FALSE)
}))

knitr::kable(kendall_df,
  caption   = "Kendall's Tau (random subsample, n = 5000)",
  row.names = FALSE)
```

Table 11: Kendall's Tau (random subsample, n = 5000)

Pair	Tau	P_value	Decision
LOS vs Charges	0.5761	0.0000000	Reject H0
LOS vs Costs	0.5858	0.0000000	Reject H0
LOS vs BirthWt	-0.1618	0.0000007	Reject H0
Charges vs Costs	0.6708	0.0000000	Reject H0
Charges vs BirthWt	-0.1145	0.0000688	Reject H0
Costs vs BirthWt	-0.1100	0.0001329	Reject H0

5.12 Test 9: Spearman's Rank Correlation

$$\rho_s = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}$$

```
spearman_df <- do.call(rbind, apply(pair_idx, 1, function(p) {
  v1 <- num_vars[p[1]]; v2 <- num_vars[p[2]]
  x <- mydata[[v1]]; y <- mydata[[v2]]
  ok <- complete.cases(x, y)
  sp <- cor.test(x[ok], y[ok], method = "spearman", exact = FALSE)
  data.frame(Pair      = paste(v1, "vs", v2),
             Rho       = round(as.numeric(sp$estimate), 4),
             P_value   = signif(sp$p.value, 4),
             Decision  = decide(sp$p.value),
             stringsAsFactors = FALSE)
}))
```



```

}))

knitr::kable(spearman_df,
  caption = "Spearman's Rank Correlation",
  row.names = FALSE)

```

Table 12: Spearman's Rank Correlation

Pair	Rho	P_value	Decision
LOS vs Charges	0.7305	0	Reject H0
LOS vs Costs	0.7478	0	Reject H0
LOS vs BirthWt	-0.1733	0	Reject H0
Charges vs Costs	0.8589	0	Reject H0
Charges vs BirthWt	-0.1257	0	Reject H0
Costs vs BirthWt	-0.1490	0	Reject H0

```

sp_mat <- cor(mydata[, num_vars], use = "pairwise.complete.obs", method = "spearman")
knitr::kable(round(sp_mat, 4),
  caption = "Spearman Correlation Matrix")

```

Table 13: Spearman Correlation Matrix

	LOS	Charges	Costs	BirthWt
LOS	1.0000	0.7305	0.7478	-0.1733
Charges	0.7305	1.0000	0.8589	-0.1257
Costs	0.7478	0.8589	1.0000	-0.1490
BirthWt	-0.1733	-0.1257	-0.1490	1.0000

5.13 Test 10: Permutation Test (Monte Carlo)

$$T^{\text{obs}} = \tilde{X}_K - \tilde{X}_Q, \quad p = \frac{1}{B+1} \left(1 + \sum_{b=1}^B \mathbf{1}\{|T^{(b)}| \geq |T^{\text{obs}}|\} \right)$$

```

# Distribution-free two-sample permutation test, supports mean or median.
perm_two_sample <- function(x, y, B = 10000, stat = "mean", seed = 42) {
  set.seed(seed)
  obs <- if (stat == "mean") mean(x) - mean(y)
  else median(x) - median(y)
  pool <- c(x, y); n1 <- length(x)
  stats <- numeric(B)
  for (b in seq_len(B)) {

```

```

ix <- sample.int(length(pool), n1)
a <- pool[ix]; bb <- pool[-ix]
stats[b] <- if (stat == "mean") mean(a) - mean(bb) else median(a) - median(bb)
}
list(observed = obs,
     p        = (1 + sum(abs(stats) >= abs(obs))) / (B + 1),
     null     = stats)
}

```

5.13.1 Permutation Test on Total Charges (Kings vs Queens)

```

set.seed(7)
K_idx <- sample(which(mydata$Hospital.County == "Kings" & !is.na(mydata$Charges)), 3000)
Q_idx <- sample(which(mydata$Hospital.County == "Queens" & !is.na(mydata$Charges)), 3000)

p_med <- perm_two_sample(mydata$Charges[K_idx], mydata$Charges[Q_idx],
                        B = 5000, stat = "median")

perm_df <- data.frame(
  Statistic = c("Median Difference (Kings - Queens)"),
  Observed  = c(round(p_med$observed, 2)),
  P_value   = c(signif(p_med$p, 4)),
  Decision  = c(decide(p_med$p)),
  stringsAsFactors = FALSE)

knitr::kable(perm_df,
  caption    = "Permutation Test: Total Charges, Kings vs Queens (B = 5000)",
  row.names = FALSE)

```

Table 14: Permutation Test: Total Charges, Kings vs Queens
(B = 5000)

Statistic	Observed	P_value	Decision
Median Difference (Kings - Queens)	2858.07	0.0112	Reject H0

Interpretation. The median difference in Charges between Kings and Queens is significant ($p = 0.011$), so the null hypothesis of exchangeable county labels is rejected. The robust median detects a real distributional shift between the two counties even when extreme outliers (one charge exceeds \$12 million in Kings) are present in the sample.

6 Comprehensive Summary

Table 15: Comprehensive Summary of All Nonparametric Tests

Result	Test
Reject H0 for every testable variable	Runs Test (9 numeric variables)
Reject H0 for all 5 variables	Sign Test (LOS, Charges, Costs, BirthWt, Severity: K vs Q)
Reject H0 for all 5 paired comparisons	Wilcoxon Signed-Rank (LOS, Charges, Costs, BirthWt, Severity: K vs Q)
Reject H0 for all 6 comparisons	Mann-Whitney U (5 vars Kings vs Queens + Charges ED vs Non-ED)
Reject H0 for all 4 variables	KS (Kings vs Queens, 4 vars)
Reject H0 for all 4 variables	KS (ED vs Non-ED, 4 vars)
Reject H0 for all 15 testable combinations	Kruskal-Wallis (4 vars x 4 groupings)
Reject H0 for LOS, Charges, Costs	Friedman (Severity, blocked by Facility)
Reject H0 for all pairs	Kendall’s Tau (6 pairs)
Reject H0 for all pairs (rho=0.86 Charges~Costs)	Spearman’s Rho (6 pairs)
Median rejects H0 (p=0.011)	Permutation (Charges median, K vs Q)

7 Conclusion, Comparison to Existing Work, and Future Research

Ten distribution-free procedures were applied to the SPARCS 2022 hospital discharge data for Kings and Queens counties. Taken together, the evidence strongly supports the claim that:

- The ordering of the raw SPARCS records is non-random (Runs Test).
- The median Length of Stay differs between the two counties (Sign Test; $p < 2.2 \times 10^{-16}$), and the paired distributions of Length of Stay, Charges, and Costs also differ (Wilcoxon Signed-Rank).
- The independent-sample distributions of Length of Stay, Charges, and Costs differ between counties (Mann-Whitney U), and Charges differ between Emergency Department and Non-Emergency Department admissions.
- The distributions of Length of Stay, Charges, Costs, and Birth Weight differ significantly between the two counties at every level of resolution (Kolmogorov-Smirnov, Kruskal–Wallis, Permutation).
- Illness severity remains a dominant driver of Length of Stay/Charges/Costs even after controlling for facility (Friedman).
- Length of Stay, Charges, and Costs are strongly and monotonically associated (Spearman $\rho \approx 0.86$ for Charges–Costs, $\rho \approx 0.73$ for Length of Stay–Charges), while Birth Weight is only weakly related.
- The Permutation Test on Total Charges confirms a significant median difference between counties ($p = 0.011$), so the null hypothesis of exchangeable county labels is rejected. The robust median detects a real distributional shift even when extreme outliers (one charge

exceeds \$12 million in Kings) are present in the sample.

These findings corroborate Stone et al. (2022), Chazard et al. (2017), and Heslin et al. (2024), who reported large geographic and severity-driven variability in hospital utilization using SPARCS or HCUP data.

Limitations. With $N \approx 370,000$ records, even tiny effects achieve astronomical significance; a useful future extension would be to add non-parametric bootstrap confidence intervals for effect sizes (Spearman ρ , median Charges differences) to complement the p -value-based conclusions. Kendall's Tau and the Permutation Test were computed on random subsamples because exact computation on the full data is infeasible.

Future work. Planned extensions include: (i) extending the analysis to all five New York City boroughs using SPARCS 2023; (ii) adding the Jonckheere–Terpstra trend test for ordered severity levels; (iii) fitting a semi-parametric time-to-discharge model in the spirit of Li et al. (2022); and (iv) a permutation-based regression to disentangle facility, payer, and demographic effects on Charges.

8 R Code

All analyses in this report were executed in R (4.3+) using base functions only, with no external packages required for the nonparametric test statistics themselves. Key functions: `custom runs_test()`, `binom.test()` for the Sign Test, `wilcox.test(..., paired = TRUE)` for the Wilcoxon Signed-Rank Test, `wilcox.test(...)` for the Mann-Whitney U Test, `ks.test()`, `kruskal.test()`, `friedman.test()` with `reshape()`, `cor.test(..., method = "kendall")` and `cor.test(..., method = "spearman")`, and custom `perm_two_sample()` for the Permutation Test.

9 References

The full reference list – spanning methodological simulation studies on hospital Length of Stay/cost data, prior SPARCS and HCUP analyses, and permutation/bootstrap theory – is provided in the accompanying LaTeX paper `Vinay_Final_Project.pdf`, Section 9. Key sources include: Swed & Eisenhart (1943); Efron & Tibshirani (1993); Ernst (2004); Fagerland & Sandvik (2009); Qualls, Pallin, & Schuur (2010); Croux & Dehon (2010); Anhoj & Olesen (2014); Moore et al. (2017); Chazard et al. (2017); Ning et al. (2017); Bujang & Sapri (2018); Dieleman et al. (2020); Nguyen et al. (2021); Stone et al. (2022); Li et al. (2022); and Heslin et al. (2024).